6.3 Resource Provenance — Content Specification

Standard: HL7 FHIR Release 4 (v4.0.1: R4 — Mixed Normative and STU) | **Permanent URL:** https://hl7.org/fhir/R4/provenance.html
Work Group: Security Work Group | **Maturity Level:** 3 (Trial Use) | **Security Category:** Not Classified

6.3 Overview: Authenticity, Trust, and Reproducibility

This specification document reproduces the official HL7 FHIR Release 4 (v4.0.1) Provenance Resource definition. Note from the HL7 specification directory: Release 4 (v4.0.1) is the permanent R4 archival specification; the newer specification version that supersedes R4 is FHIR R5 (5.0.0), with full publication history tracked in the Directory of Published Versions.

Provenance of a resource is a record that describes entities and processes involved in producing and delivering or otherwise influencing that resource. Provenance provides a critical foundation for assessing authenticity, enabling trust, and allowing reproducibility. Provenance assertions are a form of contextual metadata and can themselves become important records with their own provenance. A provenance statement indicates clinical significance in terms of confidence in authenticity, reliability, trustworthiness, integrity, and stage in lifecycle (for example, Document Completion — whether the artifact has been legally authenticated), all of which may impact security, privacy, and trust policies.

Specification Attribute	FHIR R4 Value & Scope
Resource Name & Ancestor	`Provenance` — subclass of `DomainResource` (inherits `id`, `meta`, `implicitRules`, `language`, `text`, `contained`, `extension`, `modifierExtension`)
Responsible Work Group	Security Work Group — Maturity Level 3 (TU — Trial Use)
Security & Compartments	Security Category: Not Classified | Compartments: Device, Patient, Practitioner, RelatedPerson
Referenced By Resources	Contract, DeviceRequest, MedicationAdministration, MedicationDispense, MedicationRequest, ServiceRequest, and Task

6.3.1 Scope and Usage: Context Assertions vs. Event Auditing

The `Provenance` resource tracks information about the activity that created, revised, deleted, or signed a version of a resource, describing the entities and agents involved. This information can be used to form assessments about its quality, reliability, or trustworthiness, or to provide pointers for where to go to further investigate the origins of the resource and the information in it.

Provenance resources are a record-keeping assertion that gathers information about the context in which the information in a resource was obtained. Provenance resources are prepared by the application that initiates the create/update etc. of the resource. An AuditEvent resource contains overlapping information, but is created as events occur, to track and audit the events. AuditEvent resources are often (though not exclusively) created by the application responding to the read/query/create/update/etc. event.

6.3.2 Boundaries and Relationships: Native Elements First

Many other FHIR resources contain some elements that represent information about how the resource was obtained, and therefore they overlap with the functionality of the Provenance resource. These properties in other resources should always be used in preference to the Provenance resource, and the Provenance resource should be used where additional information is required, or an explicit record of provenance is desired.

The relationship between a resource and its provenance is established by a reference from the provenance resource to its target (Provenance.target). In this way, provenance may be provided about any resource or version, including past versions. There may be multiple provenance records for a given resource or version of a resource.

6.3.3 Background and Context: W3C Provenance & ISO/HL7 Model

The Provenance resource is based on the W3C Provenance Specification, and mappings are provided. The Provenance resource is tailored to fit the FHIR use-cases for provenance more directly. In terms of W3C Provenance, the FHIR Provenance resource covers **"Generation" of "Entity"** with respect to FHIR-defined resources for creation or updating; whereas AuditEvent covers **"Usage" of "Entity"** and all other **"Activity"** as defined in W3C Provenance.

The W3C Provenance Specification defines three fundamental model concepts:

- **Entity:** An entity is a physical, digital, conceptual, or other kind of thing with some fixed aspects; entities may be real or imaginary.
- **Agent:** An agent is something that bears some form of responsibility for an activity taking place, for the existence of an entity, or for another agent's activity.
- **Activity:** An activity is something that occurs over a time period and acts upon or with entities. It may include consuming, processing, transforming, modifying, relocating, using, or generating entities.

The Provenance resource corresponds to a single activity that identifies a set of resources (target) generated by the activity. The activity also references other entities (entity) that were used and the agents (agent) that were associated with the activity. To record multiple activities that resulted in one target, record each activity in independent Provenance records all pointing at that target.

The Provenance resource depends upon having References to all the resources, entities, and agents involved in the activity. These References need not be resolvable, but they must provide a unique and unambiguous identification. If a resource, entity, or agent can have different versions that must be identified, then the Reference must have versioning information included. Versioning and unique

identification are not mandated for all systems that provide Resources, entities, and agents, though inclusion of Provenance requirements may introduce requirements for versioning and unique identification on those systems.

The Provenance resource leverages the W3C Provenance specification to represent HL7 support of provenance throughout its standards and is explicitly modeled as functional capabilities in **ISO/HL7 10781 EHR System Functional Model Release 2** and **ISO 21089 Trusted End-to-End Information Flows**.

6.3.4 Resource Content: Formal Structure & Element Definitions

The table below defines the complete hierarchical structure of the Provenance resource in FHIR R4, including flags (TU = Trial Use, Σ = Summary element), cardinality, data types, and normative bindings.

Element Path	Flags	Card.	Type	Description & Constraints
Provenance	TU	0..*	DomainResource	Who, What, When for a set of resources. Ancestor elements: `id`, `meta`, `implicitRules`, `language`, `text`, `contained`, `extension`, `modifierExtension`.
`.target`	Σ	1..*	Reference(Any)	Target Reference(s) (usually version specific).
`.occurred[x]`	—	0..1	choice	When the activity occurred (one of `occurredPeriod` or `occurredDateTime`).
`.occurredPeriod`	—	0..1	Period	Time period during which the activity occurred.
`.occurredDateTime`	—	0..1	dateTime	Point in time or dateTime when the activity occurred.
`.recorded`	Σ	1..1	instant	When the activity was recorded / updated.
`.policy`	—	0..*	uri	Policy or plan the activity was defined by.
`.location`	—	0..1	Reference(Location)	Where the activity occurred, if relevant.
`.reason`	—	0..*	CodeableConcept	Reason the activity is occurring. Binding: v3.PurposeOfUse (Extensible).
`.activity`	—	0..1	CodeableConcept	Activity that occurred. Binding: Provenance activity type (Extensible).

Element Path	Flags	Card.	Type	Description & Constraints
.agent	—	1..*	BackboneElement	Actor involved in the activity.
.agent.type	Σ	0..1	CodeableConcept	How the agent participated. Binding: Provenance participant type (Extensible).
.agent.role	—	0..*	CodeableConcept	What the agents role was. Binding: SecurityRoleType (Example).
.agent.who	Σ	1..1	Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization)	Who participated in the activity.
.agent.onBehalfOf	—	0..1	Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization)	Who the agent is representing.
.entity	—	0..*	BackboneElement	An entity used in this activity.
.entity.role	Σ	1..1	code	derivation | revision | quotation | source | removal. Binding: ProvenanceEntityRole (Required).
.entity.what	Σ	1..1	Reference(Any)	Identity of entity.
.entity.agent	—	0..*	see Provenance.agent	Entity is attributed to this agent (uses identical structure to Provenance.agent).
.signature	—	0..*	Signature	Signature on target.

6.3.4 Resource Serialization Templates (XML, JSON & Turtle)

Standard wire representations for Provenance are provided below in XML, JSON, and RDF Turtle formats as defined in the FHIR R4 specification.

XML Template

```
<Provenance xmlns="http://hl7.org/fhir">
  <!-- from Resource: id, meta, implicitRules, and language -->
  <!-- from DomainResource: text, contained, extension, and modifierExtension -->
  <target><!-- 1..* Reference(Any) Target Reference(s) (usually version specific) -->
</target>
  <occurred[x]><!-- 0..1 Period|dateTime When the activity occurred --></occurred[x]>
```

```
  <recorded value="[instant]"/><!-- 1..1 When the activity was recorded / updated -->
  <policy value="[uri]"/><!-- 0..* Policy or plan the activity was defined by -->
  <location><!-- 0..1 Reference(Location) Where the activity occurred, if relevant --
></location>
  <reason><!-- 0..* CodeableConcept Reason the activity is occurring --></reason>
  <activity><!-- 0..1 CodeableConcept Activity that occurred --></activity>
  <agent> <!-- 1..* Actor involved -->
    <type><!-- 0..1 CodeableConcept How the agent participated --></type>
    <role><!-- 0..* CodeableConcept What the agents role was --></role>
    <who><!-- 1..1
Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|Device|Organization) --
></who>
    <onBehalfOf><!-- 0..1
Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|Device|Organization) --
></onBehalfOf>
  </agent>
  <entity> <!-- 0..* An entity used in this activity -->
    <role value="[code]"/><!-- 1..1 derivation | revision | quotation | source |
removal -->
    <what><!-- 1..1 Reference(Any) Identity of entity --></what>
    <agent><!-- 0..* Content as for Provenance.agent: Entity is attributed to this
agent --></agent>
  </entity>
  <signature><!-- 0..* Signature Signature on target --></signature>
</Provenance>
```

JSON Template

```
{
  "resourceType" : "Provenance",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "target" : [{ Reference(Any) }], // R! Target Reference(s) (usually version
specific)
  // occurred[x]: When the activity occurred. One of these 2:
  "occurredPeriod" : { Period },
  "occurredDateTime" : "<dateTime>",
  "recorded" : "<instant>", // R! When the activity was recorded / updated
  "policy" : ["<uri>"], // Policy or plan the activity was defined by
  "location" : { Reference(Location) }, // Where the activity occurred, if relevant
  "reason" : [{ CodeableConcept }], // Reason the activity is occurring
  "activity" : { CodeableConcept }, // Activity that occurred
  "agent" : [{ // R! Actor involved
    "type" : { CodeableConcept }, // How the agent participated
    "role" : [{ CodeableConcept }], // What the agents role was
    "who" : {
Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|Device|Organization) },
// R!
    "onBehalfOf" : {
Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|Device|Organization) }
  }],
  "entity" : [{ // An entity used in this activity
    "role" : "<code>", // R! derivation | revision | quotation | source | removal
```

```
    "what" : { Reference(Any) }, // R! Identity of entity
    "agent" : [{ Content as for Provenance.agent }] // Entity is attributed to this
agent
  }],
  "signature" : [{ Signature }] // Signature on target
}
```

Turtle (RDF) Template

```
@prefix fhir: <http://hl7.org/fhir/> .
[ a fhir:Provenance;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root
  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Provenance.target [ Reference(Any) ], ... ; # 1..* Target Reference(s)
  # Provenance.occurred[x] : 0..1 When the activity occurred. One of these 2
    fhir:Provenance.occurredPeriod [ Period ]
    fhir:Provenance.occurredDateTime [ dateTime ]
  fhir:Provenance.recorded [ instant ]; # 1..1 When the activity was recorded /
updated
  fhir:Provenance.policy [ uri ], ... ; # 0..* Policy or plan the activity was
defined by
  fhir:Provenance.location [ Reference(Location) ]; # 0..1 Where the activity
occurred
  fhir:Provenance.reason [ CodeableConcept ], ... ; # 0..* Reason the activity is
occurring
  fhir:Provenance.activity [ CodeableConcept ]; # 0..1 Activity that occurred
  fhir:Provenance.agent [ # 1..* Actor involved
    fhir:Provenance.agent.type [ CodeableConcept ]; # 0..1 How the agent participated
    fhir:Provenance.agent.role [ CodeableConcept ], ... ; # 0..* What the agents role
was
    fhir:Provenance.agent.who [
Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|Device|Organization) ];
    fhir:Provenance.agent.onBehalfOf [
Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|Device|Organization) ];
  ], ...;
  fhir:Provenance.entity [ # 0..* An entity used in this activity
    fhir:Provenance.entity.role [ code ]; # 1..1 derivation | revision | quotation |
source | removal
    fhir:Provenance.entity.what [ Reference(Any) ]; # 1..1 Identity of entity
    fhir:Provenance.entity.agent [ See Provenance.agent ], ... ; # 0..* Entity
attributed to agent
  ], ...;
  fhir:Provenance.signature [ Signature ], ... ; # 0..* Signature on target
]
```

Changes Since FHIR Release 3 (STU3 to R4 Diff)

Between FHIR Release 3 (STU3) and Release 4 (v4.0.1), the Provenance resource underwent targeted structural updates to align data types with CodeableConcept and standardize references. All 5 R3-to-R4 conversion map tests execute cleanly with 0 errors.

Element	Modification in FHIR Release 4
`Provenance.occurred[x]`	Renamed from `period` to `occurred[x]`; added type `dateTime` alongside `Period`.
`Provenance.reason`	Type changed from `Coding` to `CodeableConcept`.
`Provenance.activity`	Type changed from `Coding` to `CodeableConcept`.
`Provenance.agent.type`	Added new element (`0..1 CodeableConcept`).
`Provenance.agent.role`	Removed extensible binding to `http://hl7.org/fhir/ValueSet/security-role-type` (now Example binding).
`Provenance.agent.who`	Renamed from `who[x]` to `who`; removed type `uri`.
`Provenance.agent.onBehalfOf`	Renamed from `onBehalfOf[x]` to `onBehalfOf`; removed type `uri`.
`Provenance.entity.role`	Changed value set URI from `http://hl7.org/fhir/ValueSet/provenance-entity-role` to versioned `http://hl7.org/fhir/ValueSet/provenance-entity-role|4.0.1`.
`Provenance.entity.what`	Renamed from `what[x]` to `what`; removed types `uri` and `Identifier`.
`Provenance.agent.relatedAgentType`	Deleted element in R4.

6.3.4.1 Terminology Bindings

The following terminology value sets are bound to coded elements within the `Provenance` resource.

Path	Definition	Binding Strength	Value Set Reference
`Provenance.reason`	The reason the activity took place.	Extensible	v3.PurposeOfUse
`Provenance.activity`	The activity that took place.	Extensible	ProvenanceActivityType
`Provenance.agent.type`	The type of participation that a provenance agent played with respect to the activity.	Extensible	ProvenanceParticipantType
`Provenance.agent.role`	The role that a provenance agent played with respect to the activity.	Example	SecurityRoleType

Path	Definition	Binding Strength	Value Set Reference
`Provenance.entity.role`	How an entity was used in an activity.	Required	ProvenanceEntityRole

6.3.4.2 – 6.3.4.5 Implementation & Operational Guidance

6.3.4.2 Using the Provenance Resource

The `Provenance` resource identifies information about another resource (the reference element). It may be used in two primary architectural patterns:

- **As part of a Document Bundle:** Identifies the provenance of part or all of the FHIR document. References are often not explicitly versioned, but they always implicitly pertain to the version of the resource found in the document.
- **On a RESTful System:** Keeps track of provenance information relating to resources over RESTful interactions. Here, the target resource reference should be version-specific. For new resources that need a corresponding `Provenance` resource, the version-specific reference is often not knowable until after the target resource has been updated. To avoid integrity problems where the target resource is updated without its provenance, the target resource and the `Provenance` resource should be submitted as a pair using a transaction.

6.3.4.2.1 HTTP Header (`X-Provenance`)

The custom HTTP header `X-Provenance` can provide a provenance resource inline when performing `PUT` or `POST` operations using the RESTful interface:

```
POST [base]/Observation
Content-Type: application/fhir+?
X-Provenance: { "resourceType": "Provenance", "location": { "reference": "Location/1"
}, "agent": ... }

[body]
```

The intent is that the server picks up the provenance payload, fills out `Provenance.target`, and stores the provenance information as it normally would. Operational constraints:

- Irrespective of the `Content-Type` or `Accept` headers, the `Provenance` resource in the header is always in JSON format.
- The header payload **SHALL NOT** have a specified `Provenance.target`. The server fills the `target` in as it processes the contents of the `POST`/`PUT` and determines the IDs of the resource(s) to which the interaction applies.
- Servers **MAY** ignore the header, but **SHOULD** process it if they support provenance.
- Some server frameworks impose a length limit on individual HTTP headers or headers as a whole; servers **MAY** reject a request if the `X-Provenance` header is too long.

- The use of the header is not documented for other HTTP methods (GET, etc.).

6.3.4.3 Signatures

The Provenance resource includes a signature element (digital signature) which can be used for standards-based integrity verification and non-repudiation purposes. The Signature datatype provides details on use of the signature element. The Signature.type coded value of "Source" should be used when the signature is simply proving that the resource content is the same as it was when the resource was updated or created.

6.3.4.4 Provenance of Removal

A Provenance record can be recorded to indicate who deleted a Resource. If versioning is supported, the version that was deleted is referenced in Provenance.target; if versioning is not supported, then Provenance.target contains the non-version reference. Provenance.entity is not used unless there is a business requirement to do so.

6.3.4.5 Use of Provenance to Record Import and Transform

Provenance can be used to record activities of an automaton that transforms input — such as middleware that extracts information from an HL7 v2 message or an HL7 CDA document and creates FHIR resources. In these cases:

- The middleware creates the target resources and a Provenance resource listing all target resources in Provenance.target.
- The middleware is identified as one of the Provenance.agent elements, with Provenance.agent.role set to assembler.
- The middleware may record the upstream source as another Provenance.agent element.
- The original content is optionally saved (e.g., as a DocumentReference or Binary), and Provenance.entity points at this original content.
- Any original source provenance covering history prior to the import transformation should be converted into FHIR Provenance records as appropriate.

6.3.5 Search Parameters

The table below lists all standard RESTful search parameters defined for Provenance (in addition to common parameters; see FHIR Searching).

Name	Type	Description	FHIRPath Expression & Target Types
agent	reference	Who participated	Provenance.agent.who (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
agent-role	token	What the agents role was	Provenance.agent.role
agent-type	token	How the agent participated	Provenance.agent.type

Name	Type	Description	FHIRPath Expression & Target Types
`entity`	reference	Identity of entity	`Provenance.entity.what` (Any)
`location`	reference	Where the activity occurred, if relevant	`Provenance.location` (Location)
`patient`	reference	Target Reference(s) (usually version specific)	`Provenance.target.where(resolve() is Patient)` (Patient)
`recorded`	date	When the activity was recorded / updated	`Provenance.recorded`
`signature-type`	token	Indication of the reason the entity signed the object(s)	`Provenance.signature.type`
`target`	reference	Target Reference(s) (usually version specific)	`Provenance.target` (Any)
`when`	date	When the activity occurred	`(Provenance.occurred as dateTime)`